QUANTITATIVE METHODS 1

Worksheet 1 – Presenting and organising data

Detailed solutions

Exercise 1 – Detailed solution

Target population: all customers who made a purchase in September. Sample: the 80 surveyed customers. Statistical unit: one customer.

Variable	Type	Suitable graph
Payment method	nominal qualitative	bar chart / pie chart
Number of items	discrete quantitative	stick/bar diagram
Time in store	continuous quantitative	histogram
Satisfaction	ordinal qualitative	ordered bars

Cumulative frequencies are meaningful for ordered variables: number of items, time and satisfaction. They are not meaningful for payment method because nominal categories have no natural order.

A large sample may still be biased if selection is not representative. For example, a voluntary survey may attract more highly satisfied or highly engaged customers.

Exercise 2 – Detailed solution

$$N = 3 + 7 + 12 + 10 + 6 + 2 = 40.$$

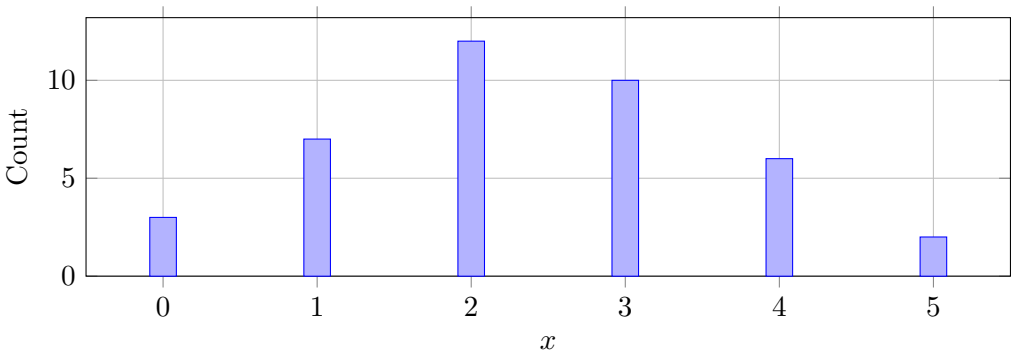
x_i	0	1	2	3	4	5
n_i	3	7	12	10	6	2
f_i	0.075	0.175	0.300	0.250	0.150	0.050
F_i	0.075	0.250	0.550	0.800	0.950	1.000

For example:

$$f_2 = \frac{12}{40} = 0.30 = 30\%.$$

The proportion with at most three orders is:

$$F(3) = \frac{3 + 7 + 12 + 10}{40} = \frac{32}{40} = 0.80 = 80\%.$$



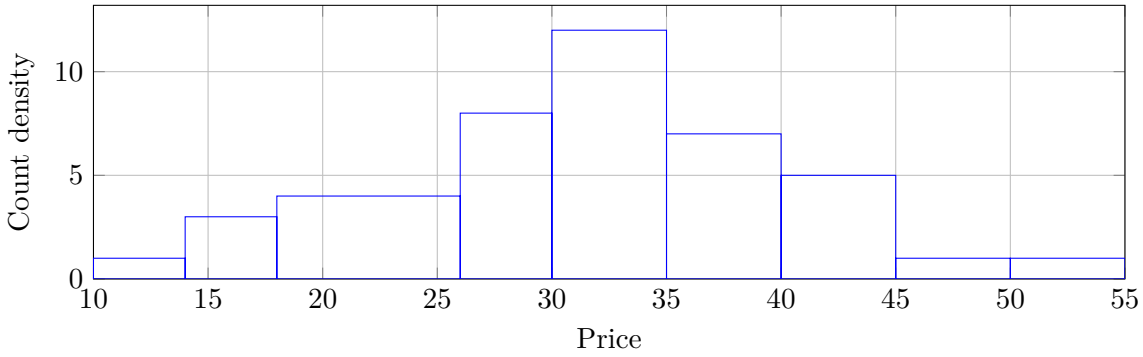
Exercise 3 – Detailed solution

Class	$[10, 14[$	$[14, 18[$	$[18, 26[$	$[26, 30[$	$[30, 35[$	$[35, 40[$	$[40, 45[$	$[45, 50[$	$[50, 55[$
n_i	4	12	32	32	60	35	25	5	5
a_i	4	4	8	4	5	5	5	5	5
$f_i(\%)$	1.90	5.71	15.24	15.24	28.57	16.67	11.90	2.38	2.38
d_i	1	3	4	8	12	7	5	1	1
ICC	4	16	48	80	140	175	200	205	210
DCC	210	206	194	162	130	70	35	10	5

For example:

$$f_{[18,26[} = \frac{32}{210} \approx 0.1524 = 15.24\%, \qquad d_{[18,26[} = \frac{32}{8} = 4.$$

With unequal class widths, rectangle area must represent the count, so density is used as height.



Less than 35 euros:

$$140, \qquad \frac{140}{210} = 66.67\%.$$

At least 35 euros:

$$70, \qquad \frac{70}{210} = 33.33\%.$$